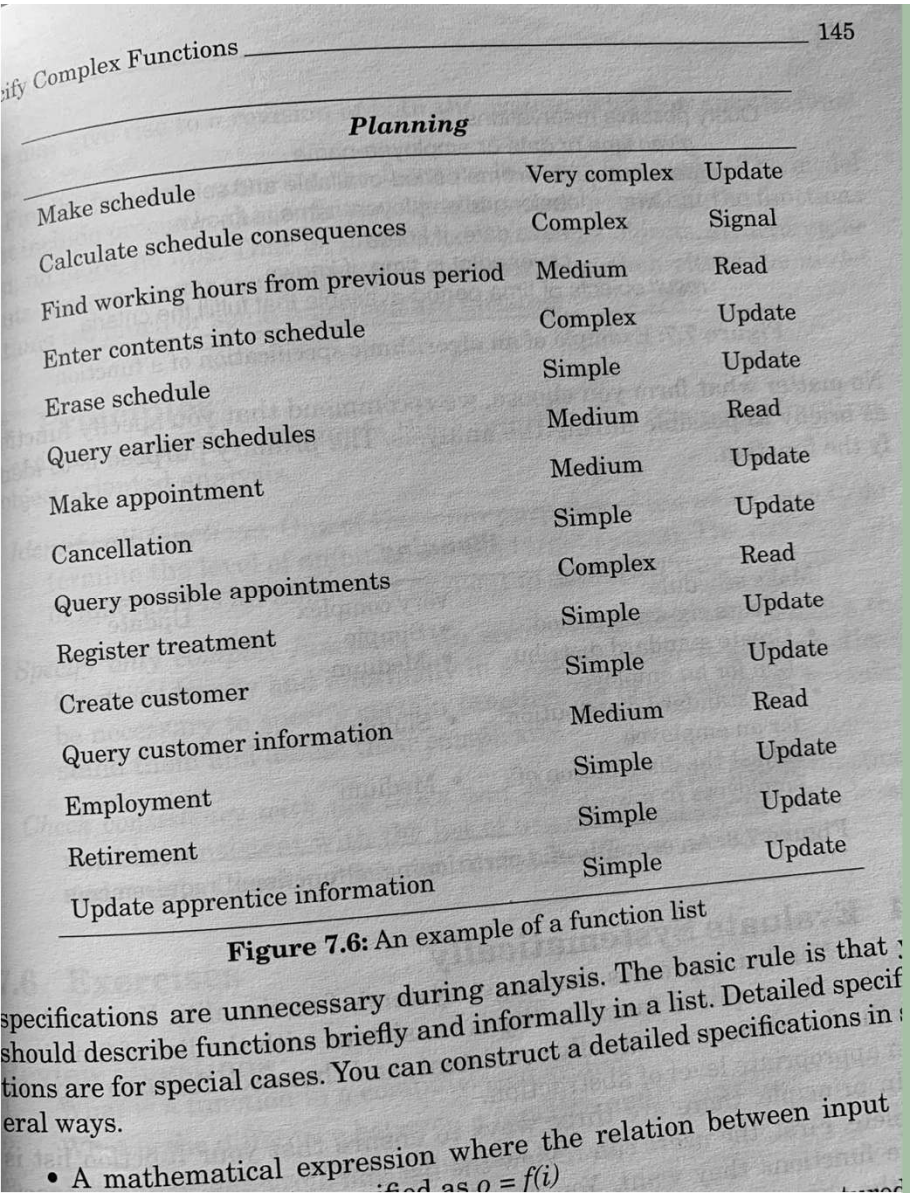


ify Complex Functions 145

| Planning | | |
|---|---|---|
| Make schedule | Very complex | Update |
| Calculate schedule consequences | Complex | Signal |
| Find working hours from previous period | Medium | Read |
| Enter contents into schedule | Complex | Update |
| Erase schedule | Simple | Update |
| Query earlier schedules | Medium | Read |
| Make appointment | Medium | Update |
| Cancellation | Simple | Update |
| Query possible appointments | Complex | Read |
| Register treatment | Simple | Update |
| Create customer | Simple | Update |
| Query customer information | Medium | Read |
| Employment | Simple | Update |
| Retirement | Simple | Update |
| Update apprentice information | Simple | Update |

**Figure 7.6:** An example of a function list

specifications are unnecessary during analysis. The basic rule is that
should describe functions briefly and informally in a list. Detailed specif
tions are for special cases. You can construct a detailed specifications in
eral ways.

- A mathematical expression where the relation between input
  ified as $o = f(i)$

# System Development

Session 7 – Application Domain: Functions

Anders Bruun & Ashna Zada

# Function

A facility for making a model useful for actors

# Identify Functions – Exercise 7.1

## Do something

In your project groups, spend 30 minutes to a) Identify functions for your semester project application and b) Divide your functions into categories

## Shoot

Make a list of your identified functions

## Examples

Functions determine what the system must do so actors may interact with your model of the Problem Domain, e.g. "Make schedule", "Make course", "Edit lecture topic" etc...

# Show the Loot - Exercise 7.1

## Share

Upload your list of functions to Miro, frame “session 7”.

## Discuss

We have some secret questions for discussion


Questions

# Questions - Exercise 7.1

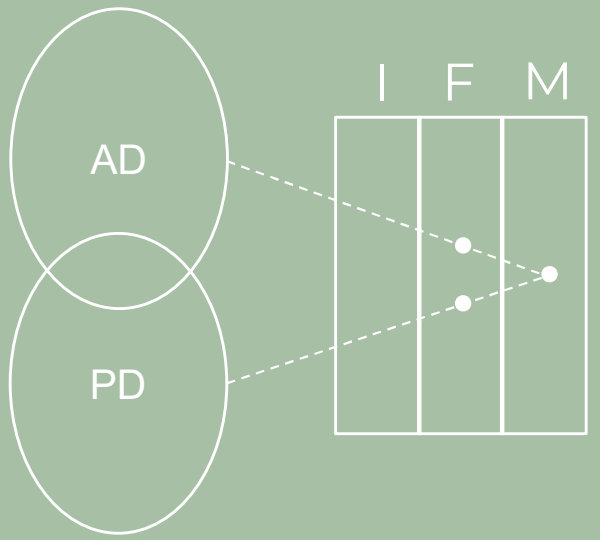
## Approach

Which of the previous outputs did you consider to identify functions? Why?

## Function Types

What different types of functions did you identify? Any differences?

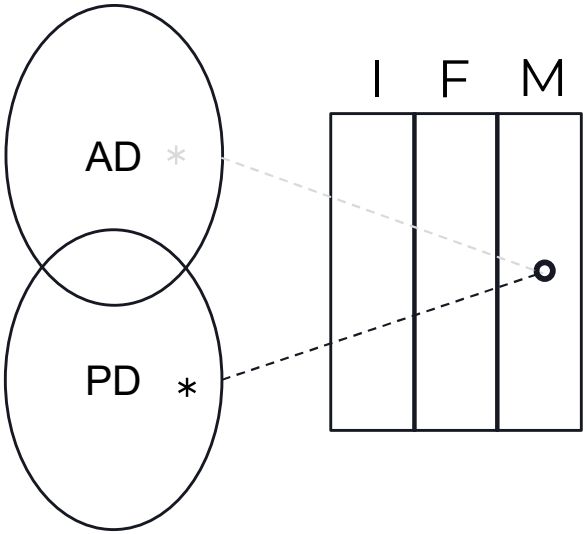
# Function Types




Express the relation between Problem Domain model and Application Domain:

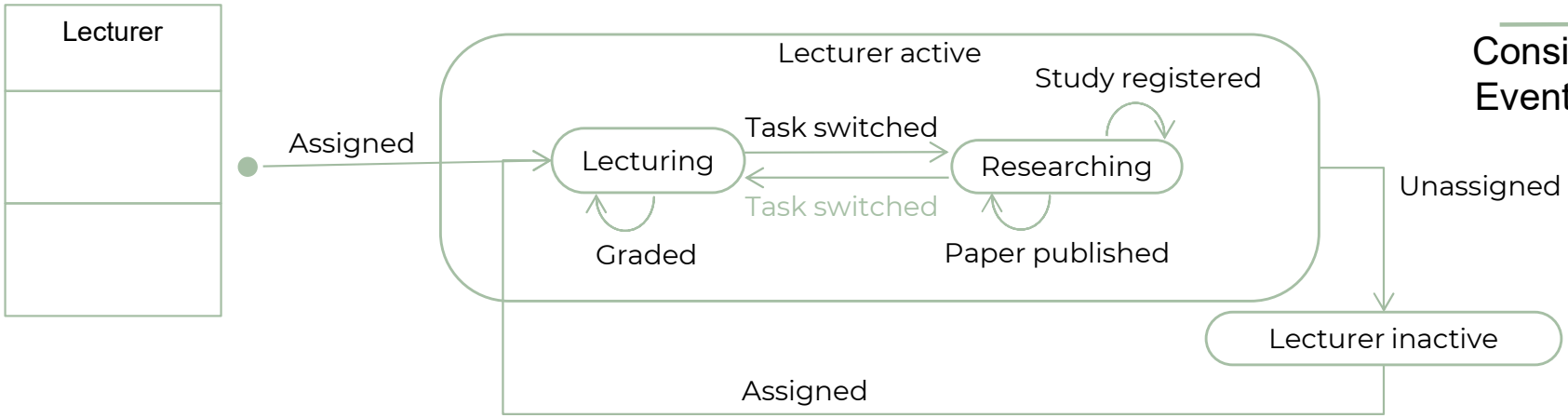
- **Update**. Function that is activated by a Problem Domain event and result in a change in the model's state
- **Signal**. Function that is activated by a change in the model's state and result in a reaction (display for an actor or intervention in the Problem Domain)
- **Read**. Function that is activated by a need for information by an actor
- **Compute**. Function that consists of computation involving information provided by an actor or the model

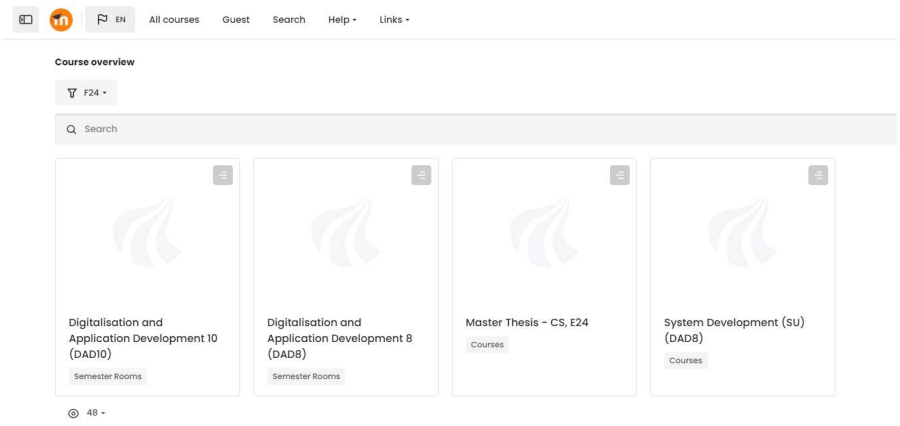
Functions are often a mix of the above. "Pure" functions are usually simpler to implement, mixed more complex.

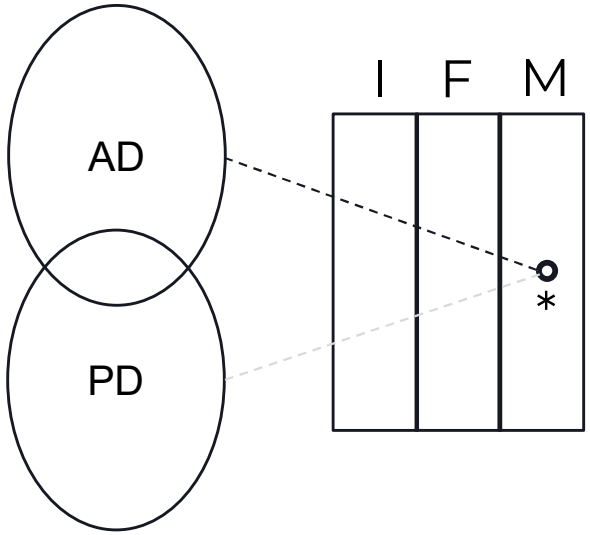



# Update example

Consider Events
Event: Assigned

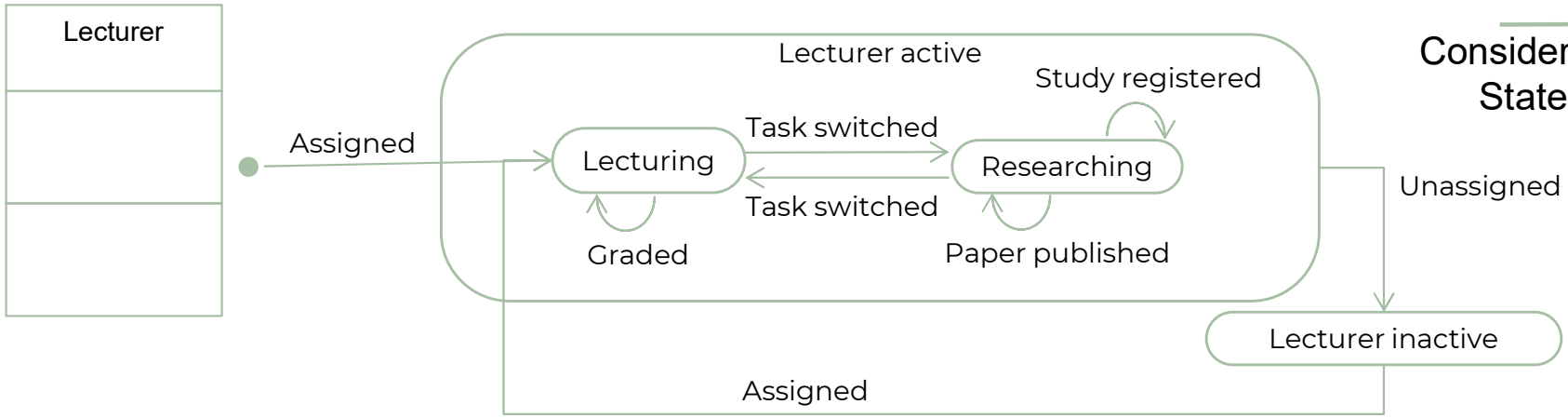


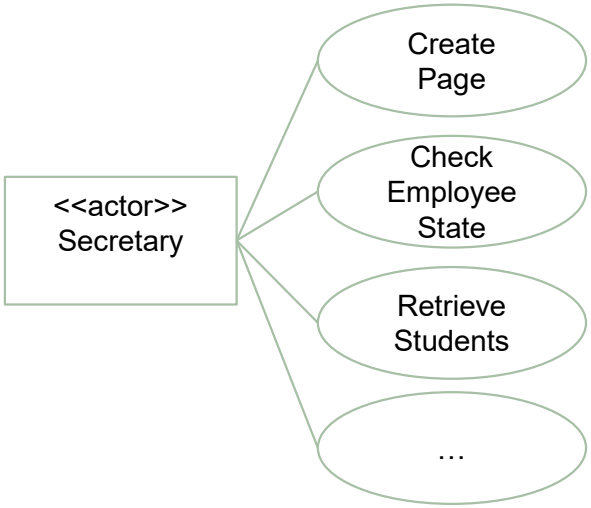


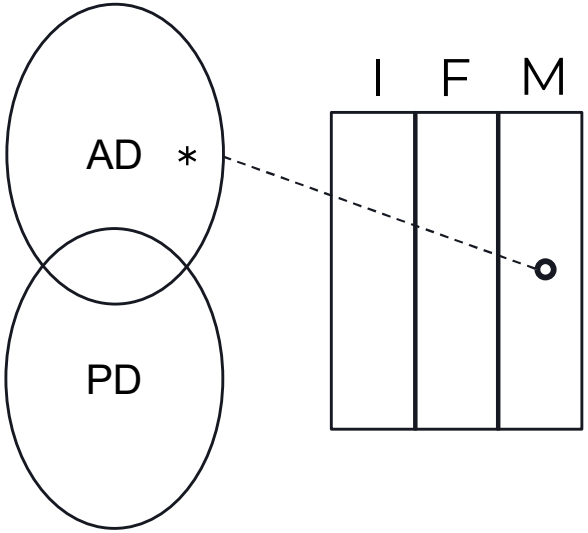




# Signal example

Consider Model States
State: Lecturing



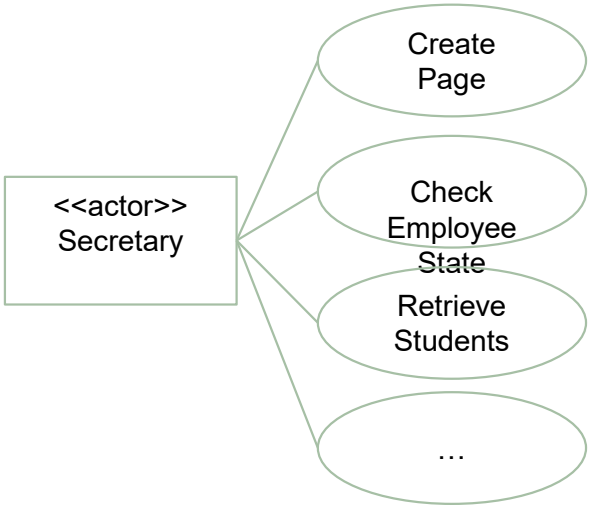


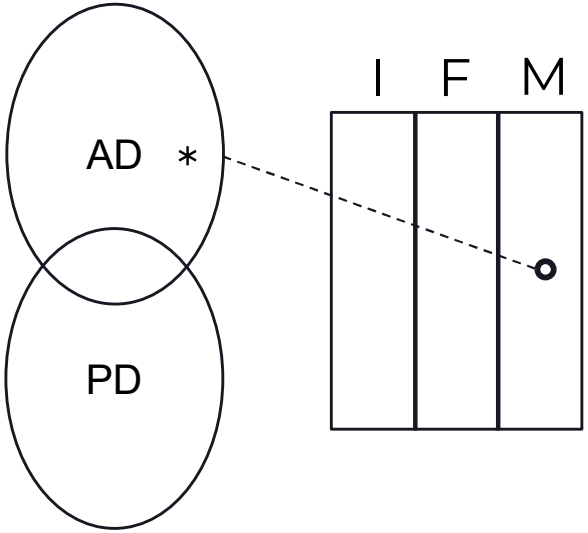




| Employee |
| --- |
| Name<br>Address<br>Phone<br>Email<br>Holiday weeks<br>Leave weeks<br>Current state |
| |

# Read example

Consider actor information needs, e.g. Lecturer state (holiday…)




# Compute example

Consider actor computational needs, e.g. find students from a given semester

# Questions when identifying functions

Some of them...

## Update

Consider events

How is the event observed and how is it registered? In which use cases does this happen?

Which objects, attributes and objects are affected by the event?

## Read

Consider actors

Given the work of the actors, what do the actors need to know about the state of the model?

Given the model, which object structures will the actors need information about?

“The source for identifying functions are partly the problem domain description, as expressed by its classes and events, and partly the application domain, as expressed by its use cases.”

-Mathiassen et al., 2000

## Compute

Consider actors and model

Which computations do the actors need to have carried out?

Does the computational basis come from the actors, model or both?

## Signal

Consider model

What are the critical states for the model?

What are the consequences when they occur?

What signals does each critical state give rise to?

# Re-Identify Functions – Exercise 7.2

## Do something

In your project groups, spend 30 minutes to identify Update, Signal, Read and Compute functions

## Shoot

Make a list of your identified functions

## Rules of Engagement

Use your semester projects as the case.

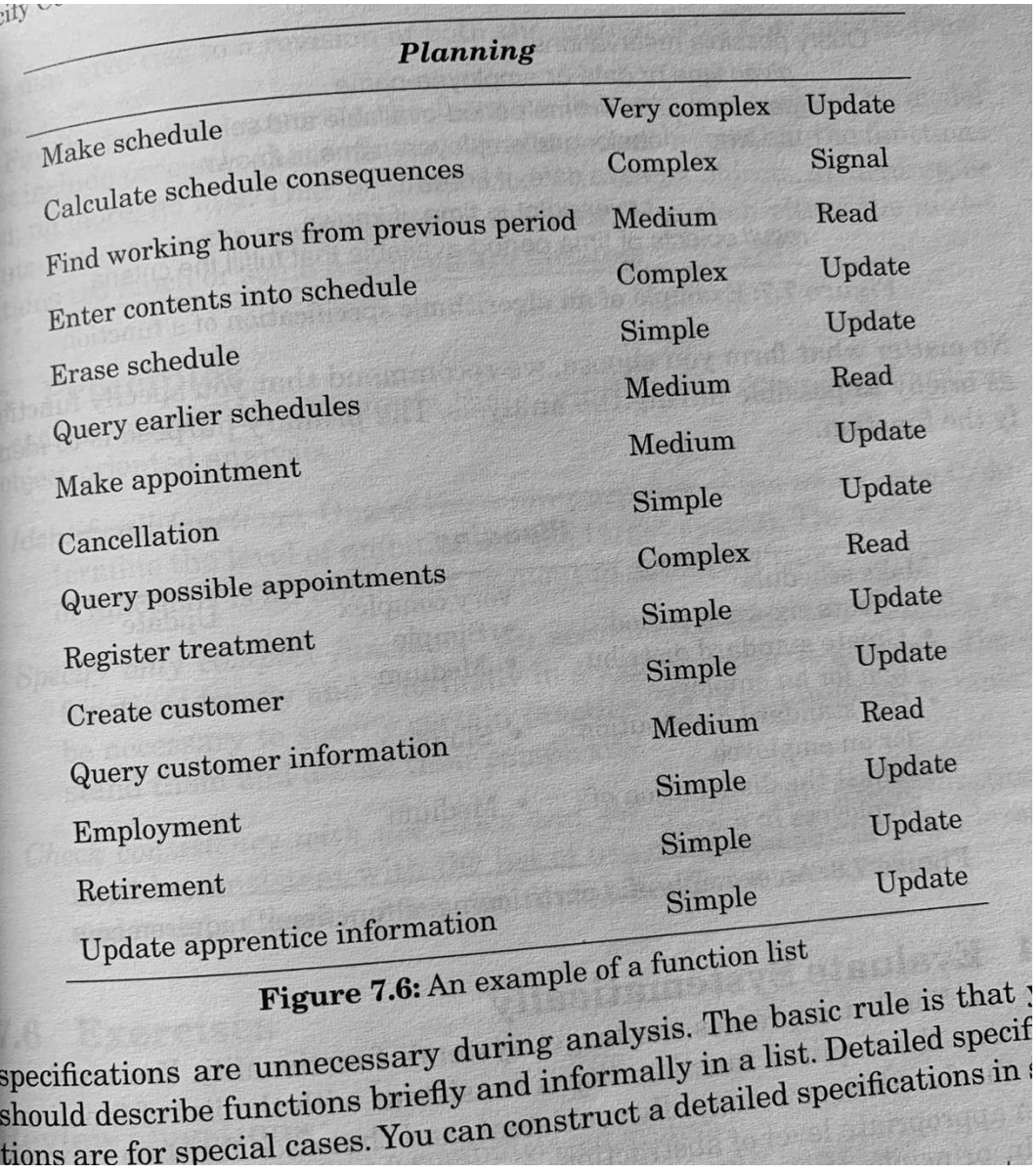
# Show the Loot - Exercise 7.2

## Share

Upload your list of functions to Miro, frame “session 7”

## Discuss

Let’s talk about the challenges you experienced



| *Planning* | | |
|---|---|---|
| Make schedule | Very complex | Update |
| Calculate schedule consequences | Complex | Signal |
| Find working hours from previous period | Medium | Read |
| Enter contents into schedule | Complex | Update |
| Erase schedule | Simple | Update |
| Query earlier schedules | Medium | Read |
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| Query possible appointments | Complex | Read |
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| Create customer | Simple | Update |
| Query customer information | Medium | Read |
| Employment | Simple | Update |
| Retirement | Simple | Update |
| Update apprentice information | Simple | Update |

**Figure 7.6:** An example of a function list

specifications are unnecessary during analysis. The basic rule is that
should describe functions briefly and informally in a list. Detailed specif
tions are for special cases. You can construct a detailed specifications in

*"Detailed specifications are for special cases... specify functions as briefly as possible during analysis."*

—Mathiassen et al., 2000

# Specifying Complex Functions

When you need a detailed specification more than a list. Three alternatives:

- **Mathematical expression**

  Output (o) as a function of input (i), *o = f(i).* E.g. a function to compute number of students in a course.

- **Pseudo Code**

  Retrieve list of students:
  if right to access
  locate all students on a course
  list students...

- **Partitioning**

  Splitting a complex function into sub-functions. E.g. sub-functions for ”Checking credentials”, ”Find students related to course” etc...

# Specify Complex Functions – Exercise 7.3

### Do something

In your project groups, spend 30 minutes to specify complex functions

### Shoot

Use one (or more) of the three types of representations to specify your complex functions (mathematical expression, pseudo code or partitioning)

### Rules of Engagement

Use your semester projects as the case

# Show the Loot - Exercise 7.3

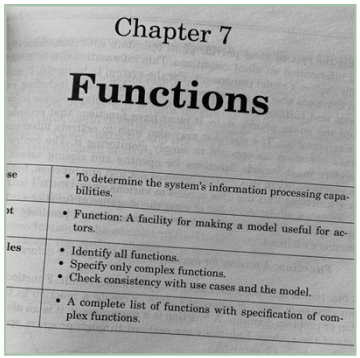
## Share

Upload your specifications to Miro, frame “session 7”.

## Discuss

Let’s talk about the challenges you experienced

# Until we meet again



Chapter 7

**Functions**

| | |
|---|---|
| se | • To determine the system's information processing capabilities. |
| ot | • Function: A facility for making a model useful for actors. |
| les | • Identify all functions.<br>• Specify only complex functions.<br>• Check consistency with use cases and the model. |
| | • A complete list of functions with specification of complex functions. |





Video series 6
Mini videos 3,4,5

# Complete Function List – Exercise 7.4

## Do something

Re-re-identify your functions and assess their complexity in terms of how complicated it will be to implement. Specify all of your “complex” / “very complex” functions

## Shoot

Make a list of all your functions, their estimated complexity and their type. Accompany this list with specifications of complex functions

## Rules of Engagement

Use your semester projects as the case

# Show the Loot - Exercise 7.4

## Share

Upload your list and specifications to Miro, frame “session 7”.

## Discuss

Let’s talk about the challenges you experienced